

```
In[1]:= Clear[pathHome, pathFGV, path]
pathHome = "G:\\Meu Drive\\ArtigoWikipedia\\PesquisaWikipedia\\";
pathFGV =
  "D:\\Users\\leandro.gorno\\Dropbox\\Research\\Projects\\2023-11 - Wikipedia
  graphs\\2025-07 - New data gathering\\Data\\";
path = pathHome;
```

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In[5]:= Clear[graphPT3]
graphPT3 = Get[path <> "graphPT3.dat"]
```

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Out[6]= Graph[ Vertex count: 25 092
Edge count: 55 474]
```

```
In[7]:= Length[EdgeList[graphPT3]]
```

```
Out[7]= 55 474
```

```
In[8]:= Length[VertexList[graphPT3]]
```

```
Out[8]= 25 092
```

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In[28]:= (*1. Gender API Function*)ClearAll[getGenero];
getGenero[titulo_String] := Module[{apiWiki, dadosWiki, pagina, qid,
  apiWD, dadosWD, claims, generoQ, apiLabel, dadosLabel}, Quiet[apiWiki =
  "https://pt.wikipedia.org/w/api.php?action=query&prop=pageprops&titles=" <>
  URLEncode[titulo] <> "&format=json";
  dadosWiki = Import[apiWiki, "RawJSON"];
  pagina = First@Values@dadosWiki["query", "pages"];
  qid = pagina["pageprops", "wikibase_item"];
  If[qid === Missing["KeyAbsent", _], Return[None]];
  apiWD = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
  qid <> "&props=claims&format=json";
  dadosWD = Import[apiWD, "RawJSON"];
  claims = dadosWD["entities", qid, "claims"];
  If[! KeyExistsQ[claims, "P21"], Return[None]];
  generoQ = claims["P21"][[1, "mainsnak", "datavalue", "value", "id"]];
  apiLabel = "https://www.wikidata.org/w/api.php?action=wbgetentities&ids=" <>
  generoQ <> "&props=labels&languages=pt&format=json";
  dadosLabel = Import[apiLabel, "RawJSON"];
  dadosLabel["entities", generoQ, "labels", "pt", "value"]]]

(*2. Extract the raw Top 20 data first*)
getRawTop20[metric_] := Take[ReverseSortBy[
  Transpose[{VertexList[graphPT3], N[metric[graphPT3]]}], Last], UpTo[20]];

rawDeg = getRawTop20[DegreeCentrality];
rawBet = getRawTop20[BetweennessCentrality];
rawEig = getRawTop20[EigenvectorCentrality];

(*3. Find unique articles and fetch their
  genders ONCE (to save time and avoid API bans)*)
Print["Consultando Wikidata... isso pode levar alguns segundos."];
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uniqueArticles = Union[Join[rawDeg[[All, 1]], rawBet[[All, 1]], rawEig[[All, 1]]];
genderDictionary = AssociationMap[getGenero, uniqueArticles];

(*4. Row Formatter: color por género y formato numérico por métrica*)
ClearAll[formatRow];
formatRow[{article_, score_}, fmt_] := {If[genderDictionary[article] === "feminino",
  Style[article, Bold, Darker[Red]], article], fmt[score]};

fmtDeg = Function[s,
  NumberForm[Round[s], DigitBlock → 3, NumberSeparator → ".", NumberPoint → ","]];
fmtBet = Function[s, NumberForm[s / 10^6, {6, 2}, NumberPoint → ","]];
fmtEig = Function[s, NumberForm[s, {7, 5}, NumberPoint → ","]];

(*5. Apply the formatting to our raw lists*)
topDeg = TableForm[Map[formatRow[#, fmtDeg] &, rawDeg], TableAlignments → {Left, Right}];
topBet = TableForm[Map[formatRow[#, fmtBet] &, rawBet], TableAlignments → {Left, Right}];
topEig = TableForm[Map[formatRow[#, fmtEig] &, rawEig], TableAlignments → {Left, Right}];

(*6. Build the Grid, forcing exactly equal widths*)
Tablacentralidad =
Grid[{{Item[Style["Centralidad de Grado", Bold], ItemSize → Scaled[1 / 3],
  Alignment → Center], Item[Style["Centralidad de Intermediación", Bold],
  ItemSize → Scaled[1 / 3], Alignment → Center],
  Item[Style["Centralidad de vector propio", Bold], ItemSize → Scaled[1 / 3],
  Alignment → Center]}}, {Item[topDeg, Alignment → Center],
  Item[topBet, Alignment → Center], Item[topEig, Alignment → Center]}}, Frame → All,
Background → {{Lighter[Blue, 0.95], Lighter[Green, 0.95], Lighter[Orange, 0.95]},
{Lighter[Gray, 0.8], None}}, Alignment → Top]

(*7. Exportar los valores crudos,
sin formato ni redondeo*) Export["centralidad_raw.csv",
Join[{{"metrica", "articulo", "valor"}}, Map[{"grado", #[[1]], #[[2]]} &, rawDeg],
Map[{"intermediacion", #[[1]], #[[2]]} &, rawBet], Map[{"vector", #[[1]], #[[2]]} &, rawEig]]]
Consultando Wikidata... isso pode levar alguns segundos.

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Out[45]=

Centralidad de Grado	Centralidad de Intermediación	Centralidad de vector propio			
iz Inácio Lula da Silva	1.2	Úlio Vargas	20	Úlio Vargas	0,00
Úlio Vargas	1.0	Chado de Assis	15	z Inácio Lula da Silva	0,00
William Shakespeare	89	etano Veloso	14	co Buarque	0,00
boleão Bonaparte	82	iz Inácio Lula da Silva	14	nando Henrique Cardoso	0,00
ernando Henrique Cardoso	78	rge Amado	10	tano Veloso	0,00
us	67	boleão Bonaparte	10	berto Gil	0,00
abel II do Reino Unido	63	co Buarque	10	é Sarney	0,00
etano Veloso	56	n-Paul Sartre	9	nando Collor	0,00
ico Buarque	56	berto Gil	9	el Brizola	0,00
é Sarney	53	us	9	icius de Moraes	0,00
atão	52	é Sarney	8	a Bethânia	0,00
lberto Gil	51	abel II do Reino Unido	7	Jobim	0,00
ernando Collor	50	nando Henrique Cardoso	7	e Amado	0,00
onna	48	William Shakespeare	6	celino Kubitschek	0,00
Chado de Assis	45	atão	6	o Gilberto	0,00
rge Amado	34	ia Fagundes Telles	5	ld de Andrade	0,00
lson Rodrigues	32	e Guevara	5	e Serra	0,00
onel Brizola	32	z Fernando Carvalho	5	redo Neves	0,00
ernando Pessoa	32	s de Camões	5	onio Carlos Magalhães	0,00
n Jobim	29	onna	5	o de Andrade	0,00

Out[46]=

centralidad_raw.csv